



HIGH-PERFORMANCE DATA SUITE
POSTGRES SQL // POWER BI // SYNTHETIC TELEMETRY // FULL-STACK ANALYTICS

PRODUCTION SPEC V4.2

LIVE APEX TELEMETRY

FORMULA 1 TELEMETRY & RACE ANALYTICS PLATFORM

An end-to-end telemetry engineering and computational intelligence ecosystem. Spanning advanced PostgreSQL window functions, multi-dimensional Power BI star schema modeling with DAX measures, and an 8,000-row synthetic multi-parameter telemetry feature engine.

PILLAR 01 // DATABASE ENGINE

SQL ENGINE

PostgreSQL Analytical Layer

8 mission-critical SQL analytical queries targeting `formula_one_data`: CTE window partitioning, driver points aggregation, circuit records, overtake gainers, and pit crew variance.

Core: CTEs • PARTITION BY • STDDEV • GROUP BY

PILLAR 02 // VISUAL BI SYSTEM

POWER BI & DAX

Executive Telemetry Cockpit

High-refresh Power BI dashboard architecture utilizing Star Schema modeling, dynamic DAX measures (Podium %, Win Rates, Net Grid Gain), clustered visuals, and interactive race telemetry.

Core: CALCULATE • DIVIDE • Star Schema • PBIX

PILLAR 03 // DATA ENGINEERING

BIG DATA PIPELINE

Synthetic Telemetry Engine

High-cardinality synthetic telemetry dataset comprising 8,000 granular rows x 58 engineered features: tyre thermal degradation, sector splits, track dynamics, and weather vectors.

Core: 8,000 Laps • 58 Features • Tyre Wear • Sectors

SYSTEM ARCHITECTURE

END-TO-END TECHNOLOGY STACK & DATA PIPELINE

5 ARCHITECTURAL TIERS

TIER 01 // SYNTHESIS

Python & Pandas

DATA GENERATOR

High-entropy telemetry generation scripts with sensor noise injection, delta times, and wear curves.

- Python 3.11 & NumPy
- openpyxl & Tabular CSV
- 58 Synthetic Columns

TIER 02 // PERSISTENCE

PostgreSQL 16

RELATIONAL RDBMS

Normalized tabular data storage optimized for complex window partitioning and aggregation runs.

- ANSI SQL Windowing
- CTE Rank Partitioning
- Statistical Aggregations

TIER 03 // MODELING

Power BI Engine

ENTERPRISE BI

Star Schema data transformation engine with 1-to-many dimensional relationships and Power Query ETL.

- Star Schema Architecture
- Dim / Fact Partitioning
- Automated Relationships

TIER 04 // CALCULATIONS

DAX Measure Core

DYNAMIC METRICS

Dynamic context transition logic calculating pole conversions, finish percentages, and pit variances.

- Row & Filter Contexts
- Safe Division (DIVIDE)
- Filter Overrides (CALCULATE)

TIER 05 // VISUALS

Cockpit UI & Web

EXECUTIVE CANVAS

Dual presentation layers: Power BI Clustered Visuals & HTML5 / Tailwind live telemetry dashboards.

- Interactive Visual Filtering
- Chart.js Web Telemetry
- Executive PDF Snapshots

PIPELINE THROUGHPUT SPECIFICATION

Seamless Cross-Stack Interoperability

Raw telemetry ingested as tabular records, processed via PostgreSQL analytical windows, and sliced in real-time through compiled DAX expressions.

8,000

DATASET ROWS

58

TELEMETRY PARAMS

< 15ms

SQL QUERY LATENCY

POSTGRESQL ADVANCED RACE ANALYTICS ENGINE

8 ANALYTICAL QUERIES // CTEs & WINDOWING

ANALYTICAL QUERY CATALOG (`FORMULA\_ONE\_PGRES.SQL`)

POSTGRESQL 16

ID	ANALYTICAL OBJECTIVE	CORE LOGIC / SQL METHOD	KEY OUTPUT
Q1	Longest Circuit by Track Distance	ORDER BY circuit_length_km DESC LIMIT 5	Spa: 7.004 km
Q2	Championship Points Titans (All-time)	SUM(points) GROUP BY driver, nationality	Hamilton: 4.8k+
Q3	Fastest Lap Time Hall of Fame	ORDER BY fastest_lap_time_sec ASC (DNF=0)	Monza (1:18.88)
Q4	Season Consistency King (Window Function)	ROW_NUMBER() OVER(PARTITION BY season)	Verstappen (9.88)
Q6	Overtake Master (Net Grid Positions Gained)	ROUND(AVG(grid_gain), 2) HAVING COUNT > 15	Alonso (+3.59/gp)
Q7	Pole-to-Win Conversion Rate (%)	SUM(CASE WHEN grid=1 AND pos=1) / COUNT	Monaco: 73.3%
Q8	Pit Crew Reliability & Standard Deviation	AVG(dur) & STDDEV(dur) HAVING stops >= 20	RBR: 2.28s (±0.34)

Q4 WINDOW PARTITIONING ARCHITECTURE

CTE PATTERN

```
WITH ranked_drivers AS ( SELECT season, driver, nationality,
ROUND(AVG(consistency_score::numeric), 2) AS avg_score, ROW_NUMBER() OVER(
PARTITION BY season ORDER BY AVG(consistency_score) DESC ) AS rnk FROM
formula_one_data GROUP BY season, driver, nationality HAVING COUNT(round) >= 5
) SELECT * FROM ranked_drivers WHERE rnk = 1;
```

Q8 PIT CREW VARIANCE & RELIABILITY

STDDEV STATS

```
SELECT constructor, COUNT(*) AS stops,
ROUND(AVG(average_pit_duration)::numeric, 2) AS avg_dur,
ROUND(STDDEV(average_pit_duration)::numeric, 2) AS pit_stddev FROM
formula_one_data WHERE average_pit_duration > 0 AND average_pit_duration < 40
GROUP BY constructor HAVING COUNT(*) >= 20;
```

POWER BI TELEMETRY COCKPIT & DAX MODELING

• STAR SCHEMA // COMPILED DAX EXPRESSIONS

DIMENSIONAL MODELING ARCHITECTURE

Power BI Star Schema Implementation

Engineered from raw flat telemetry into an enterprise-ready dimensional star schema to optimize VertiPaq engine compression and cross-filtering speed:

Dim_Driver: DriverID (PK), Driver Name, Nationality, Racing Number	Dim_Circuit: CircuitID (PK), Track Name, Country, Length_KM, Turns
Dim_Season: SeasonID (PK), Year, Total Rounds, Regulation Era	Fact_RaceResults: ResultID, DriverFK, CircuitFK, Grid, Finish, DNF, Pts

VISUAL CONTAINER BREAKDOWN (`FORMULA ONE DASHBOARD.PBIX`)

Clustered Bar Career Points Titans	Combo Chart Wins & Poles / Season	Donut Chart Podium Conversion %
Column Chart Circuit Length Ranked	Info Matrix Pit Stop StdDev & Consistency	KPI Cards DNF % & Total Races

DAX MEASURE // POLE-TO-WIN CONVERSION %

SAFE DIVISION

```
Pole to Win % = VAR PolesFinished = CALCULATE(COUNTROWS(formula_one_data), formula_one_data[grid_position] = 1, formula_one_data[dnf] = 0) VAR PoleAndWon = CALCULATE(COUNTROWS(formula_one_data), formula_one_data[grid_position] = 1, formula_one_data[finish_position] = 1, formula_one_data[dnf] = 0) RETURN DIVIDE(PoleAndWon, PolesFinished, 0) * 100
```

DAX MEASURE // PIT VARIANCE (STANDARD DEVIATION)

STDEV.S

```
Pit Variance StdDev = CALCULATE( STDEV.S(formula_one_data[average_pit_duration]), formula_one_data[average_pit_duration] > 0, formula_one_data[average_pit_duration] < 40 )
```

DAX MEASURE // PODIUM CONVERSION RATE

FILTER CONTEXT

```
Podium Rate % = VAR Podiums = CALCULATE(COUNTROWS(formula_one_data), formula_one_data[finish_position] <= 3, formula_one_data[dnf] = 0) VAR Starts = CALCULATE(COUNTROWS(formula_one_data), formula_one_data[dnf] = 0) RETURN DIVIDE(Podiums, Starts, 0) * 100
```

SYNTHETIC TELEMETRY ENGINEERING (8,000 ROWS X 58 PARAMETERS)

HIGH-CARDINALITY FEATURE ENGINEERING

DOMAIN 01 // VEHICLE & ENGINE

Aerodynamics & Powertrain

High-frequency physics parameters evaluating downforce trim, ICE thermal health, and electrical recovery:

- speed_trap_kmh
- throttle_pct, brake_bar
- gear_selection, rpm_avg
- ers_deployment_mj, drs_active

DOMAIN 02 // LAP & SECTORS

Micro-Sector Timing Traces

Sector splits and apex delta tracking to isolate cornering efficiency versus straight-line speed:

- sector_1_time_sec
- sector_2_time_sec
- sector_3_time_sec
- delta_to_fastest_lap
- theoretical_optimal_lap

DOMAIN 03 // STRATEGY & TYRES

Tyre Wear & Pit Durations

Pirelli compound degradation decay curves, carcass surface temperatures, and pit-lane ingress:

- tyre_compound (Soft/Med/Hard)
- tyre_age_laps, tyre_wear_pct
- tyre_surface_temp_celsius
- average_pit_duration
- pit_stop_count, undercut_delta

DOMAIN 04 // ENVIRONMENT

Weather & Track Grip Evolution

Ambient atmospheric inputs and dynamic track rubbering-in coefficient across race stints:

- air_temperature_celsius
- track_temperature_celsius
- humidity_pct, air_density
- rainfall_intensity_mm
- track_rubbering_state

DOMAIN 05 // RACE DYNAMICS

Incidents & Grid Movement

Neutralizations, overtaking deltas, race pace stabilization, and classified retirements:

- grid_position, finish_position
- grid_gain (Overtake Delta)
- dnf (Did Not Finish Flag)
- safety_car_laps, vsc_laps
- gap_to_leader_sec

DOMAIN 06 // MODELING METRICS

Algorithmic Scoring Indices

Engineered indices serving as synthetic ground-truth targets for predictive models:

- consistency_score (0-10)
- aggressive_overtake_index
- tyre_preservation_rating
- pit_crew_efficiency_score
- championship_points_won

FILE: synthetic_f1_ultimate_analytics_dataset.csv

ENCODING: UTF-8 // COMMA-DELIMITED // 8,000 ROWS

INTEGRITY: 100% COMPLETE // NO SENSOR DROPOUTS

PRODUCTION DELIVERY & TECHNOLOGY COMPARATIVE MATRIX

• ARCHITECTURAL COMPARISON & ROADMAP

TECHNOLOGY LAYER	CORE TOOLING	ANALYTICAL ROLE & PROCESSING MODEL	LATENCY & SCALE PROFILE	KEY BUSINESS / RACING DELIVERABLE
Data Ingestion	Python, Pandas, openpyxl	Synthesizes 58 telemetry features; enforces realistic physics, weather noise, and tyre degradation curves.	Batch processing: 8,000 rows generated in ~1.2s.	GROUND-TRUTH TELEMETRY ENGINE
Relational OLAP	PostgreSQL 16, ANSI SQL	Executes complex multi-table aggregations, season window partitioning (Q4), and pit standard deviation (Q8).	Index scans < 15ms across all 8 benchmark queries.	STANDARDIZED RACING RECORDS CATALOG
BI Semantic Model	Power BI Desktop, DAX	Translates SQL logic into dynamic filter context measures (Pole-to-Win %, Net Grid Gain, Podium %).	Sub-second in-memory columnar compression (VertiPaq).	EXECUTIVE STRATEGY DASHBOARD
Web Telemetry	HTML5, Tailwind, Chart.js	Provides responsive dark-mode telemetry cockpit with live season filters and embedded SQL Query Inspector.	Instant client-side rendering (0ms server latency).	INTERACTIVE RACE ENGINEER HUD

DATA GOVERNANCE & ACCURACY

Zero Missing Values Policy

Queries strictly guard against DNF contamination (**dnf** = 0) and erroneous pit durations (**0** < **dur** < **40s**) to ensure clean metric reporting.

MODULAR EXTENSIBILITY

Plug-and-Play Strategy AI

The 58-feature telemetry output is pre-normalized for training scikit-learn or XGBoost tyre life prediction models and pit stop undercut classifiers.

EXECUTIVE READY

Dual Presentation Paradigms

Serves both technical race engineers via raw SQL inspections and team principals through executive Power BI and PDF visual slide decks.